

Riemann Roch Generalized

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Rapid fire concepts we know:

1. We shall take varieties and schemes for granted
2. We shall be working with algebraic curves, usually algebraic plane curves; hence curves given by $f(x, y)$ for coordinate functions x, y .
3. Such curves are projective ([1]). If they are smooth, we naturally ask about their relation to one-dimensional complex manifolds; these are called Riemann surfaces (as they are 2 real dimensional)
4. Compact Riemann surfaces are classified up to homeomorphism by their genus, and $\chi(X) = 2 - 2g$ where χ is the Euler Characteristic.
5. Via normalization, we have the following (This is in fact proven in [2])

Theorem 0.0.1: Characterizing Riemann Surfaces Using Normalization

Any compact Riemann surface $\tilde{C}$ can be obtained through the normalization of a certain plane algebraic curve C with at most ordinary double points, i.e. There exists a holomorphic mapping $\sigma : \tilde{C} \rightarrow \mathbb{P}^2(\mathbb{C})$ such that $\sigma(\tilde{C})$ is an algebraic curve possessing at most ordinary double points

6. The geometry polynomial function (by FTC) makes them surjective and n -to-1 except at the ramification points.
7. We shall take holomorphic and meromorphic differential forms for granted
8. Key result about holomorphic differential forms:

Theorem 0.0.2: Stokes Theorem For Holomorphic Differentials

Let C be a Riemann surface, $\Omega \subseteq C$ an open set, $\overline{\Omega}$ compact and $\partial\Omega = \gamma$ a piecewise smooth curve, and ω a holomorphic differential defined on an open set containing $\overline{\Omega}$. Then:

$$\int_{\partial\Omega} \omega = 0$$

Proof :

use local coordinates and apply Cauchy's theorem to get

$$\int_{\partial\Omega_i} \omega = 0$$

on the components, which then gluing gives the desired result

9. Residue is winding number

0.1 Poincaré-Hopf Formula

Let C be a compact Riemann surface. Then we know that for any nonconstant $f \in K(C)$ (the set of meromorphic functions on C), we get:

$$\sum_{p \in C} \nu_p(f) = 0 \quad (1)$$

that is, the number of zeros plus the number of poles (counting multiplicity) is zero. Phillip actually has this really cool generalization of the proof that I got in [3], he proves the following:

Theorem 0.1.1: Residue Theorem For Forms

Let C be a compact Riemann surface and $\omega \in K^1(C)$ a differential form (the K^1 is his notation for diff. form). Then:

$$\sum_{p \in C} \text{Res}_p(\omega) = 0$$

Proof :
exercise

Then you can deduce the usual equation (1) by setting $\omega = \frac{df}{f}$.

Bibliography

- [1] Nathanael Chwojko-Srawley. *Everything You Need To Know About Algebraic Geometry*. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algGeo.pdf.
- [2] Phillip A. Griffiths. *Introduction to Algebraic Curves*. Vol. 78. Mathematical Monographs. American Mathematical Society.
- [3] Nathanael Chwojko-Srawley. *Everything You Need To Know About Complex Analysis*. 1st ed. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_complex_analysis.pdf.